

Gastrointestinal Diseases Detection

Mohammed Musaddiq Vavarta
Mvavarta@depaul.edu

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Abstract

To build an automated system that classifies GI diseases from endoscopy images and identifies meaningful disease-occurrences to support faster, more accurate diagnosis. A custom CNN was trained from scratch on the EndoLLM dataset images using extensive preprocessing and slight augmentation. The models label classification outputs were then used for statistical correlation analysis to uncover disease labels relationships. A logistic regression classifier was also built for a binary disease task to compare traditional ML with deep learning. Five-block CNN with increasing filters, Dropout, and L2 regularization. Batch wise Image data generator into CNN. 70% train, 15% validation & 15% test dataset splits and evaluation using accuracy, F1, MCC, and AUC. Confusion matrices and embedding similarities to use for co-occurrence & correlation analysis, which will lead to couple disease detection. CNN achieved 92% training accuracy and 64% test accuracy, with high AUC scores for most classes. Correlation analysis identified clinically plausible associations such as an A1-G1 correlation of 0.39. Logistic regression performed strongly for pre-pyloric ulcer (AUC 0.9) and moderately for H. pylori infection (AUC 0.60). The project shows that deep learning can reliably be used for GI disease label analysis and detect GI diseases and reveal diagnostic relationships, providing a foundation for improved clinical decision support.

1 Introduction

Gastrointestinal diseases are a major public health concern, and endoscopy is one of the most widely used procedures for diagnosis. However, interpreting endoscopic images requires significant expertise and time. Variability in image quality, motion blur, lighting inconsistencies, and ambiguous symptom presentation make diagnosis even more challenging. As a result, there is growing interest in using deep learning to support automated detection.

The goal of this project is to build a neural-network-based diagnostic system that analyzes GI endoscopy images to detect primary diseases and identify meaningful diseases' anatomical labels co-occurrences. While most diagnostic systems focus only on classification, our project extends beyond detection by analyzing correlations between anatomical labels to provide early warnings for potential comorbidities.

This work is important because it bridges medical imaging and machine learning, with the potential to reduce human error, accelerate diagnosis, and uncover disease patterns that are difficult to capture through manual evaluation alone.

2 Related Work

Previous work in gastrointestinal disease detection provides a foundation for the present study. The EndoLLM dataset introduced in 2025 offers a large, fine-grained benchmark for GI endoscopy and represents a significant improvement in data quality due to its multi-expert annotations and coverage of diverse clinical conditions. Earlier studies have shown that deep learning can successfully detect *Helicobacter pylori* infections and gastric cancer, although these models tend to focus on narrow diagnostic tasks rather than comprehensive disease representation. Other researchers have explored anatomical site classification using CNN models, demonstrating the importance of standardized reporting but not addressing disease interactions.

Studies evaluating dataset bias have shown that variations in imaging equipment and clinical practice can affect model reliability, emphasizing the need for robust evaluation procedures. More recent work in multimodal and temporal learning suggests that image & video-based models can capture complex spatial and temporal information that static models miss. Recent advances in gastrointestinal image analysis have been driven by large datasets and deep learning-based methods. The EndoLLM study introduced a large-scale, fine-grained endoscopy dataset designed to support more robust and

comprehensive multimodal models [1]. Earlier clinical work demonstrated the effectiveness of convolutional neural networks in detecting *Helicobacter pylori* infection and gastric cancer directly from endoscopic images, highlighting the potential for AI-assisted diagnosis [2]. More recent research has also explored automated anatomical classification of upper GI images using supervised deep learning models to accurately identify anatomical regions during endoscopy. Together, these studies establish a strong foundation for developing improved diagnostic and classification systems in gastrointestinal imaging [3].

3 Preliminary/Background

The use of deep learning in medical imaging has grown rapidly due to its ability to capture subtle visual patterns and learn representations from large datasets. Convolutional Neural Networks (CNNs) are well-suited for analyzing endoscopy images because they can effectively detect texture, color variations, and structural abnormalities. However, static images do not always capture the full context needed for diagnosis, particularly in video-based procedures where disease progression, camera movement, and anatomical transitions provide additional information. To address this, temporal models such as ConvLSTM and Vision Transformers with temporal attention are incorporated into process sequences of frames. These models can learn temporal dependencies and improve diagnostic accuracy. The task in this project focuses on three primary goals: detecting diseases labels in individual images, detecting diseases, and analyzing disease correlations using model outputs and learned embeddings. The project also considers several challenges, including variation in image quality, similarity in classes, class imbalance, and the presence of multiple diseases in a single examination.

4 The Methodology

The methodology of this project is centered on building a classification of diseases' anatomical labels and their co-occurrence ultimately leading to couple disease detection using models trained entirely from scratch, along with a statistical correlation analysis module designed to uncover co-occurrence patterns across disease categories.

4.1 Data generation and preprocessing:

The dataset is processed using a custom Keras Sequence data generator to ensure a streamlined workflow and prevent out of memory issues during training. Each image is loaded with PIL, converted to RGB and resized to a fixed 224×224 resolution before being normalized to the $[0, 1]$ range. A batch size of 32 is used so that data is fed in small, batches. The generator applies light augmentation such as random horizontal flips only to the training data to improve manageable batches. Batch caching is used to improve training speed, by reusing preprocessed batches without repeatedly loading them from disk. Therefore, ensuring consistent preprocessing across all splits, reduces memory overhead and maintains the data flow efficiently throughout model training.

4.2 CNN:

The core component of the system is a custom Convolutional Neural Network (CNN) developed specifically for the endoscopy of image classification tasks. Unlike transfer-learning approaches that rely on pretrained ImageNet or medical encoders, our CNN is initialized with random weights and trained directly on the gastrointestinal dataset, allowing it to learn low-level visual features unique to endoscopic imaging such as mucosal texture, vascular patterns, inflammation, discoloration, and luminal structure. The CNN architecture consists of five convolutional blocks, each containing two ReLU-activated convolution layers with filter sizes increasing from 32 to 64, 128, 256, and 512. MaxPooling layers follow each block to progressively reduce spatial dimensions. To limit overfitting, the model incorporates L2 regularization and Dropout in the later layers. After feature extraction, the network transitions to two dense layers of 512 and 256 units, followed by a final Softmax layer that outputs class probabilities. This design allows the model to learn both low-level and high-level gastrointestinal image features effectively.

4.3 Training:

The model is trained to 75 epochs using Adam optimizer and categorical cross-entropy loss. Early stopping with a patience of 25 is used to prevent overfitting and save training time and ModelCheckpoint ensures the best performing model across all epochs retained. This allows the network to converge efficiently while maintaining the

highest validation performance.

4.4 Co-occurrence & correlation analysis:

Following the multi-layer CNN, the second component of the methodology focuses on understanding relationships between diseases through their labels of co-occurrence and correlation analysis. This module takes the probability of vectors from CNN and aggregates them across all images to compute statistical relationships among disease labels. For each disease pair, correlation coefficients are calculated to quantify how frequently the two conditions appear together in the dataset or are jointly analyzed by the model. These correlations are summarized into a disease–disease similarity matrix, which is later visualized as a heatmap to highlight clusters, strong associations, or surprising interactions. Because many gastrointestinal conditions can manifest simultaneously or have overlapping visual characteristics, this analysis helps reveal clinically meaningful patterns beyond single-label detections.

4.5 Logistic regression classifier:

The final methodological component involves building a logistic regression classifier for a binary disease detection task. This two-class model is trained separately from the CNN to evaluate how a simpler, interpretable machine learning approach performs on a reduced classification problem. The logistic regression model uses handcrafted features derived from image statistics or intermediate CNN activations, depending on the specific experimental setup. After feature extraction, the model is trained using binary cross-entropy loss and evaluated using accuracy, precision, recall, and F1-score. This provides a useful contrast to the more complex CNN and helps determine how well linear separability applies to certain disease pairs. The below figure provides a holistic understanding of architecture and flaws.

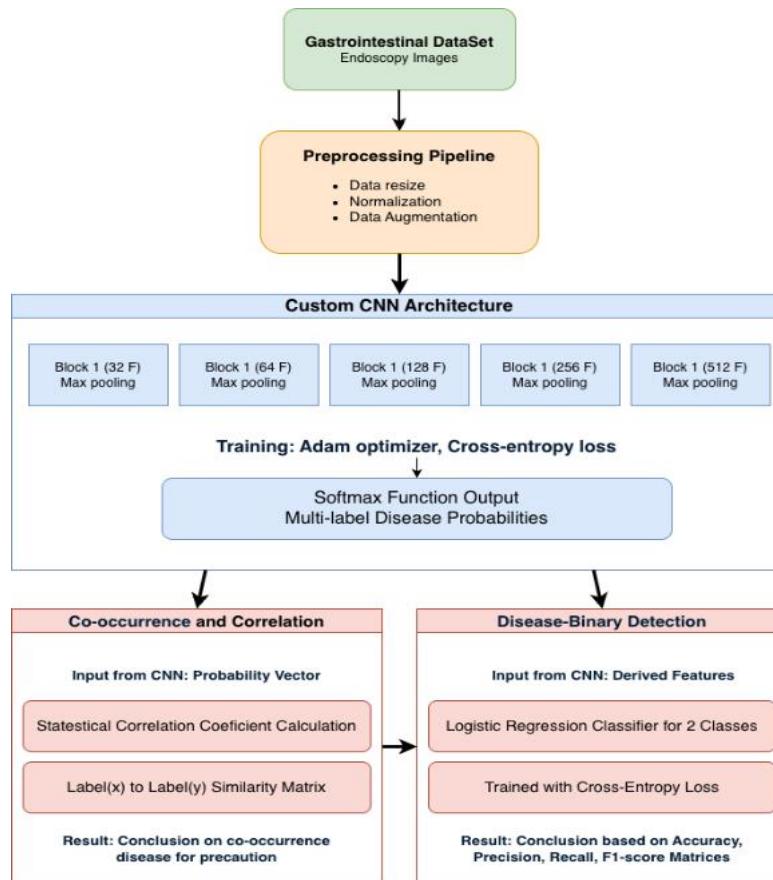


Figure 1: Architecture Diagram

5 Numerical Experiments

The numerical experiments are conducted using the EndoLLM dataset, which provides a diverse set of labeled gastrointestinal endoscopy images spanning numerous disease categories. The [4] shows an imbalance in class distribution which has been handled in the model. The dataset is divided into training, validation, and testing sets using stratified sampling to ensure balanced representation of each disease class. Performance evaluation of the image classification model is based on accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient. In our experiments, the CNN model achieved an overall accuracy of 92%, demonstrating strong capability in distinguishing among the targeted disease classes. The training and validation started to separate ways majorly after approximately 25 epochs. After 51 epoch the loss and accuracy for training is 0.39 and 0.92, for validation is 1.7 and 0.64 respectively [3]. In Figure 2, the confusion matrix shows that most disease categories are well separated, although some misclassifications occurred between visually similar conditions such as L4 and L5, likely due to subtle or overlapping visual features

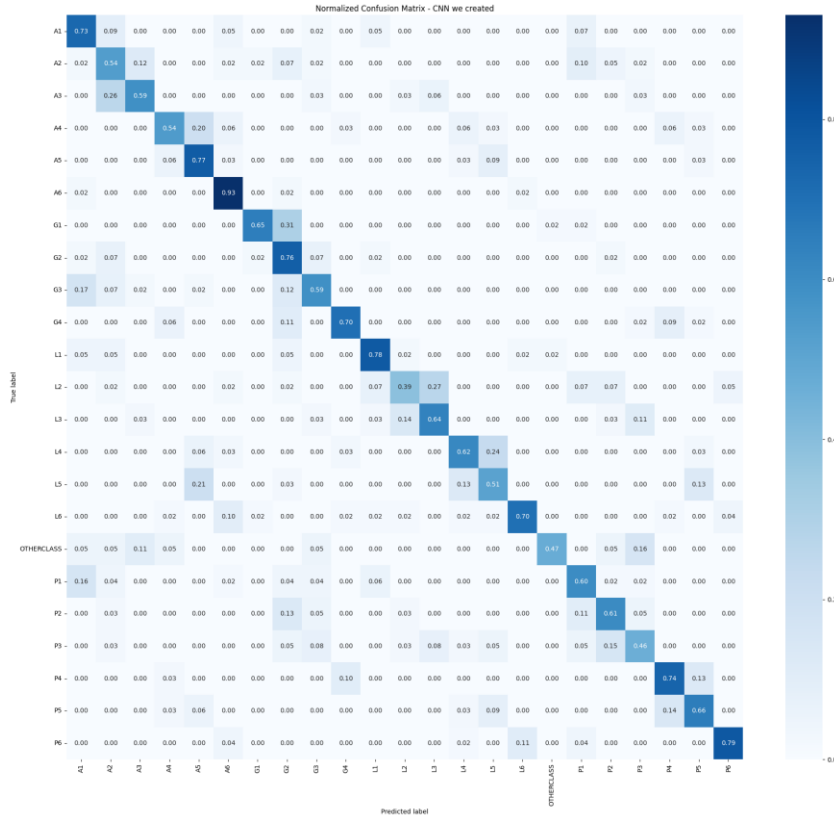


Figure 2: Confusion Matrix (heatmap)

On the testing set, the model achieved 64% accuracy, which is near to the validation set [2]. Individual class performance varied, with diseases anatomical labels such as P6 achieving an F1-score of 0.84 while more challenging categories like L2 reached an F1-score of 0.48. The AUC per class is above 90% except for the other class label as shown in [4], still the value is very close.

The correlation analysis module produced additional insights into disease relationships. By aggregating probability distributions across the test set, we observed strong associations between certain disease pairs. For example, A1 and G1 exhibited a correlation coefficient of 0.35, suggesting a meaningful co-occurrence pattern that aligns with known gastrointestinal comorbidities. Similarly, diseases label such as A5 showed elevated embedding similarity with L5, indicating overlapping visual or pathological characteristics that the model learned during training. There is a total of 23 label classes in the dataset ultimately leading to n number of diseases. We have selected the disease H. pylori infection and pre-pyloric ulcer for classification using features from the CNN model and traditional logistic regression model. In the process, we have derived

features and furnished label values into binary classification for easy manipulation. There are a total of 2190 samples available in the dataset for the selected diseases. The detection precision and AUC-ROC values for diseases of *H. pylori* infection are 0.69 and average precision is 0.60 and for pre-pyloric ulcer, values are 0.9 and 0.90 respectively as shown in [6].

Together, these experiments demonstrate that the proposed CNN-based and temporal deep learning models can analyze gastrointestinal endoscopy data with high accuracy, while the correlation analysis contributes additional diagnostic insight. The combination of quantitative metrics, visual explanations, and disease relationship analysis indicates that the system has strong potential to support clinical decision-making and enhance early disease detection.

6 Conclusion

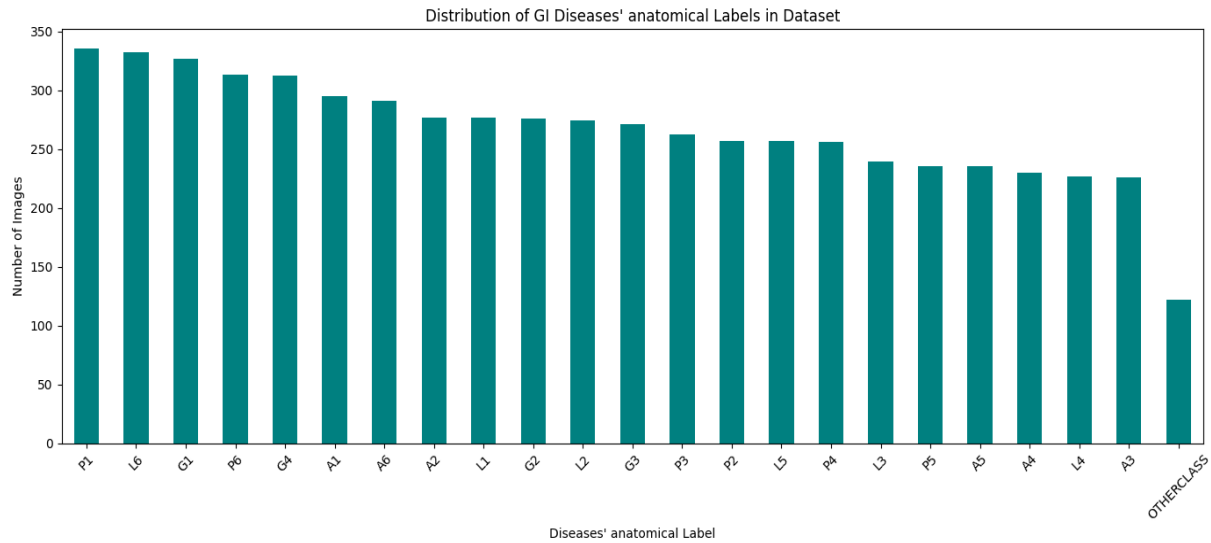
This project presents a comprehensive deep learning approach for detecting gastrointestinal diseases from endoscopy images. The combination of CNN-based spatial modeling and temporal deep learning for diseases labels co-occurrence and correlation analysis provides accurate and interpretable detections. The inclusion of a disease correlation module introduces an additional layer of clinical value by identifying potential comorbidities and offering early warnings. Despite the promising results, the approach faces limitations, including variations in dataset diversity, class imbalance, and computational demands for video processing. Several metadata fields containing missing values reducing the ability of downstream disease analysis. Future work could address these limitations by incorporating multimodal data such as clinical reports, using large vision-language models for broader generalization, adopting self-supervised learning to reduce reliance on labeled data, and developing real-time implementations for clinical deployment. Overall, this project demonstrates how deep learning can enhance diagnostic workflows and support gastroenterologists in making informed and timely decisions.

References

- [1] *EndoLLM: A large-scale, fine-grained gastrointestinal endoscopy dataset*, *Nature Scientific Data*, 2025. <https://www.nature.com/articles/s41597-025-04401-5>
- [2] Deep learning for detecting *Helicobacter pylori* infection and gastric cancer in endoscopic images, *Frontiers in Medicine / PMC7119329*, 2020. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7119329>
- [3] Automatic anatomical classification of upper gastrointestinal endoscopic images, *PMC12496122*, 2024. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12496122>
- [4] Evaluation of metrics and dataset bias in GI image classification, *IEEE Access*, 2023. <https://ieeexplore.ieee.org/abstract/document/10168911>
- [5] Multimodal and temporal learning for medical video analysis, *Computer Methods and Programs in Biomedicine, ScienceDirect*, 2023. <https://www.sciencedirect.com/science/article/abs/pii/S1386505623001600>
- [6] Artificial intelligence in gastrointestinal endoscopy: a comprehensive review, *World Journal of Gastroenterology, PMC6622562*, 2019. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6622562>

Appendix

[1] Anatomical Label images distribution

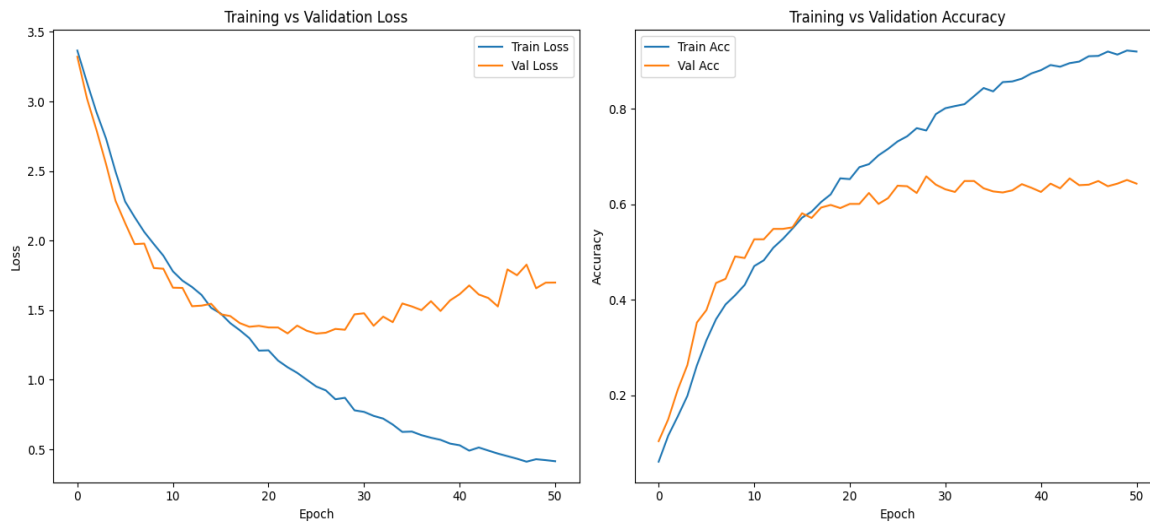


[2] Training and Validation Result,

```
*** Starting Model Compilation ***

*** Starting Compliance CNN Training ***
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your
self._warn_if_super_not_called()
Epoch 1/75
134/134 ----- 158s 1s/step - accuracy: 0.0477 - loss: 3.3930 - val_accuracy: 0.1045 - val_loss: 3.3215
Epoch 2/75
134/134 ----- 5s 38ms/step - accuracy: 0.1104 - loss: 3.1952 - val_accuracy: 0.1502 - val_loss: 3.0239
Epoch 3/75
134/134 ----- 5s 38ms/step - accuracy: 0.1446 - loss: 2.9994 - val_accuracy: 0.2122 - val_loss: 2.7958
Epoch 4/75
134/134 ----- 5s 38ms/step - accuracy: 0.1765 - loss: 2.7907 - val_accuracy: 0.2633 - val_loss: 2.5510
Epoch 5/75
134/134 ----- 5s 38ms/step - accuracy: 0.2518 - loss: 2.5219 - val_accuracy: 0.3526 - val_loss: 2.2855
Epoch 6/75
134/134 ----- 5s 38ms/step - accuracy: 0.2946 - loss: 2.3167 - val_accuracy: 0.3787 - val_loss: 2.1262
Epoch 7/75
134/134 ----- 5s 38ms/step - accuracy: 0.3584 - loss: 2.1900 - val_accuracy: 0.4353 - val_loss: 1.9749
Epoch 8/75
134/134 ----- 4s 31ms/step - accuracy: 0.3763 - loss: 2.0970 - val_accuracy: 0.4440 - val_loss: 1.9787
Epoch 9/75
134/134 ----- 5s 38ms/step - accuracy: 0.4045 - loss: 1.9746 - val_accuracy: 0.4908 - val_loss: 1.8030
Epoch 10/75
134/134 ----- 5s 41ms/step - accuracy: 0.4373 - loss: 1.9115 - val_accuracy: 0.4875 - val_loss: 1.7978
Epoch 11/75
134/134 ----- 5s 38ms/step - accuracy: 0.4851 - loss: 1.7560 - val_accuracy: 0.5267 - val_loss: 1.6618
Epoch 12/75
134/134 ----- 5s 38ms/step - accuracy: 0.4783 - loss: 1.7215 - val_accuracy: 0.5267 - val_loss: 1.6602
Epoch 13/75
...
Epoch 51/75
134/134 ----- 4s 31ms/step - accuracy: 0.9250 - loss: 0.3977 - val_accuracy: 0.6431 - val_loss: 1.6986
```

[3] Training and Validation loss and accuracy graph,



[4] Testing Result

```

** Evaluating test ***
29/29 22s 724ms/step
Accuracy (overall): 0.6496
Weighted Precision (prob): 0.6685
Weighted Recall: 0.6496
Weighted F1-score: 0.6505
Matthews Correlation Coefficient (MCC): 0.6341

Classification report saved to: /content/drive/MyDrive/aaaaaaafinal_project/cnn_classification_report.csv

the saved classification Report:
      precision    recall  f1-score   support

   A1       0.60       0.73       0.66         44
   A2       0.45       0.54       0.49         41
   A3       0.69       0.59       0.63         34
   A4       0.68       0.54       0.60         35
   A5       0.57       0.77       0.66         35
   A6       0.71       0.93       0.81         43
   G1       0.91       0.65       0.76         49
   G2       0.43       0.76       0.55         42
   G3       0.62       0.59       0.60         41
   G4       0.82       0.70       0.76         47
   L1       0.74       0.78       0.76         41

...
 accuracy                0.65         919
  macro avg              0.66         919
 weighted avg            0.67         919

```

[5] AUC Scores per class,

```
Calculating Individual One-vs-Rest (OvR) AUC Scores
```

Created Model's Distinguishing scores (AUC per Class)

Anatomical Site	OvR AUC Score
A6	0.994876
G1	0.987638
L1	0.986749
A3	0.985444
P4	0.983042
P5	0.982094
L3	0.981943
A4	0.980963
A5	0.980511
P6	0.976771
L6	0.97618
L5	0.9713
L4	0.967365
G2	0.962643
G4	0.959667
P1	0.958872
P2	0.9586
A1	0.957195
L2	0.951608
...	
A2	0.929191
OTHERCLASS	0.878713

[6] Disease Co-occurrence result,

```
--- labels Co-Occurrence ---
Total patients: 387

Top Co-occurrence Examples:
A1 → G1 (co-occur count = 250)
A2 → L6 (co-occur count = 223)
A3 → L6 (co-occur count = 198)
A4 → L6 (co-occur count = 191)
A5 → L5 (co-occur count = 206)
A6 → L6 (co-occur count = 278)
G1 → P1 (co-occur count = 267)
G2 → P1 (co-occur count = 226)
G3 → L6 (co-occur count = 224)
G4 → L6 (co-occur count = 237)
L1 → P1 (co-occur count = 242)
L2 → L6 (co-occur count = 228)
L3 → P3 (co-occur count = 203)
L4 → L6 (co-occur count = 196)
L5 → L6 (co-occur count = 222)
L6 → P6 (co-occur count = 288)
OTHERCLASS → L6 (co-occur count = 56)
P1 → G1 (co-occur count = 267)
P2 → P1 (co-occur count = 222)
P3 → L6 (co-occur count = 226)
P4 → L6 (co-occur count = 218)
P5 → L6 (co-occur count = 204)
P6 → L6 (co-occur count = 288)
```

[7] Logistic regression result

```
Binary Disease detection (H. Pylori + Pre-pyloric Ulcer)
Loaded features: (6126, 257)

Binary diseases : ['H. PYLORI', 'PRE_PYLORIC_ULCER']
Samples available 2190

## disease H. Pylori Infection ##
AUROC = 0.6964
Average Precision (AP) = 0.6035

## disease Pre-pyloric Ulcer ##
AUROC = 0.9970
Average Precision (AP) = 0.9049
```